

SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	October 1, 31	October 16
2021	October 20	October 6
2022	October 9	October 25
2023	October 28	October 14
2024	October 17	October 2
2025	October 7	October 21
2026	October 26	October 10
2027	October 15	October 29

PLANETS

- 2020: October 1** Mercury is at greatest evening elongation, magnitude 0.3.
- 2020: October 13** Mars is at opposition, magnitude -2.6.
- 2021: October 25** Mercury is at greatest morning elongation, magnitude -0.3.
- 2021: October 29** Venus is at greatest evening elongation, magnitude -4.4.
- 2022: October 8** Mercury is at greatest morning elongation, magnitude -0.3.
- 2023: October 23** Venus is at greatest morning elongation, magnitude -4.4.
- 2025: October 29** Mercury is at greatest evening elongation, magnitude 0.1.
- 2026: October 4** Saturn is at opposition, magnitude 0.4.

ECLIPSES

- 2022: October 25** A partial eclipse of the Sun is visible from Europe, northeast Africa, the Middle East, and west Asia.
- 2023: October 14** An annular eclipse of the Sun is visible from west US, southern North America, Colombia, and Brazil.
- 2023: October 28** A partial eclipse of the Moon is visible from the east of North and South America, Europe, Africa, Asia, and Australia.
- 2024: October 2** An annual eclipse of the Sun is visible from south Chile and south Argentina.

OCTOBER

The Square of Pegasus takes center stage in the northern skies in both hemispheres, a sign of the arrival of the northern fall and the southern spring. Northeast of it lies the Andromeda Galaxy, the closest large galaxy to the Earth. South of the Square, a band of faint zodiacal constellations crosses the sky, from Aries in the east to Capricornus in the southwest.

NORTHERN LATITUDES

THE STARS

The Square of Pegasus lies high in the sky from mid-northern latitudes. From one corner of the Square, the stars of Andromeda extend northeastward toward Perseus and Cassiopeia. Capella twinkles above the horizon in Auriga, lower in the northeast. In the north, the Big Dipper is at its lowest, and it is below the horizon for observers south of about latitude 30°N.

Directly beneath the Square of Pegasus is a loop of stars known as the Circlet, representing the body of one of the fish in the zodiacal constellation of Pisces. Fomalhaut (see

p.253) in Piscis Austrinus is low on the southern horizon beneath the stars of Aquarius. In the western sky, the Summer Triangle lingers, while in the east, Taurus leads the stars of winter into view.

DEEP-SKY OBJECTS

October evenings are a good time to view the Andromeda Galaxy, M31 (see pp.312-313). It can be seen as an elongated misty patch with the naked eye, if skies are not too polluted, and it is easily visible through binoculars, spanning a greater width than a full moon.

High in the north, M52, an open cluster near Cassiopeia, is visible through binoculars. Between this and the Square of Pegasus lies an often-overlooked planetary nebula, NGC 7662, nicknamed the Blue Snowball. A small telescope is needed to see it.

METEOR SHOWER

One of the year's lesser showers, the Orionids, reaches a peak of some 25 meteors an hour around October 21. They radiate from northern Orion, near the border with Gemini. This area rises late, so the meteors are best seen after midnight.

